



CLMS User Manual

Calibration Laboratory Management System

ISO/IEC 17025 · NABL Accreditation Ready

Version 2.0 · June 2026

Table of Contents

1. [Getting Started — Login & Navigation](#)
2. [Dashboard — At a Glance](#)
3. [Jobs — Creating & Managing Calibration Jobs](#)
4. [Job Workspace — Full Calibration Workflow](#)
 - a. [Datasheet Tab](#)
 - b. [Uncertainty Tab](#)
 - c. [Certificate Tab](#)
 - d. [Calibration History Tab](#)
 - e. [Remarks Tab](#)
5. [Instruments — Asset Registry](#)
6. [Customers — Client Management](#)
7. [Reference Standards — NABL Master Equipment](#)
8. [Certificates — View & Download](#)

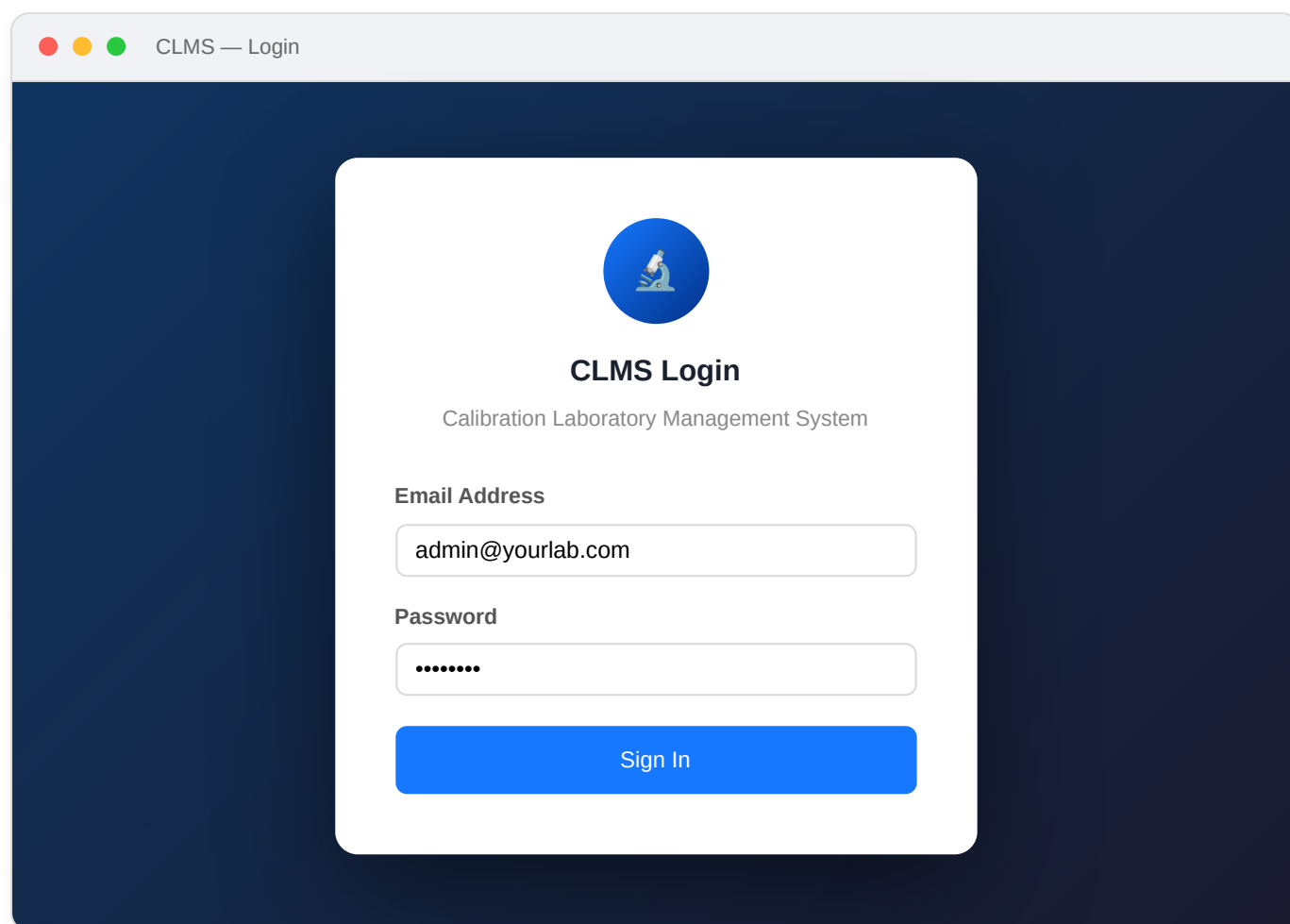
- 9. [Reports — Analytics & Exports](#)
- 10. [Settings — Lab Configuration](#)
- 11. [Customer Portal — Self-Service Access](#)
- 12. [Role Permissions Summary](#)

1 Getting Started — Login & Navigation

Access the system, understand the layout, and navigate between modules

1.1 Logging In

Open the CLMS URL in your browser. You will see the login screen.



The screenshot shows a web browser window titled "CLMS — Login". The main content area has a dark blue background. In the center, there is a white rounded rectangle containing the login form. At the top of this rectangle is a circular logo with a microscope icon. Below the logo, the text "CLMS Login" is displayed in bold, followed by "Calibration Laboratory Management System" in a smaller font. The form includes two input fields: "Email Address" with the value "admin@yourlab.com" and "Password" with masked characters ".....". A blue "Sign In" button is positioned at the bottom of the form.

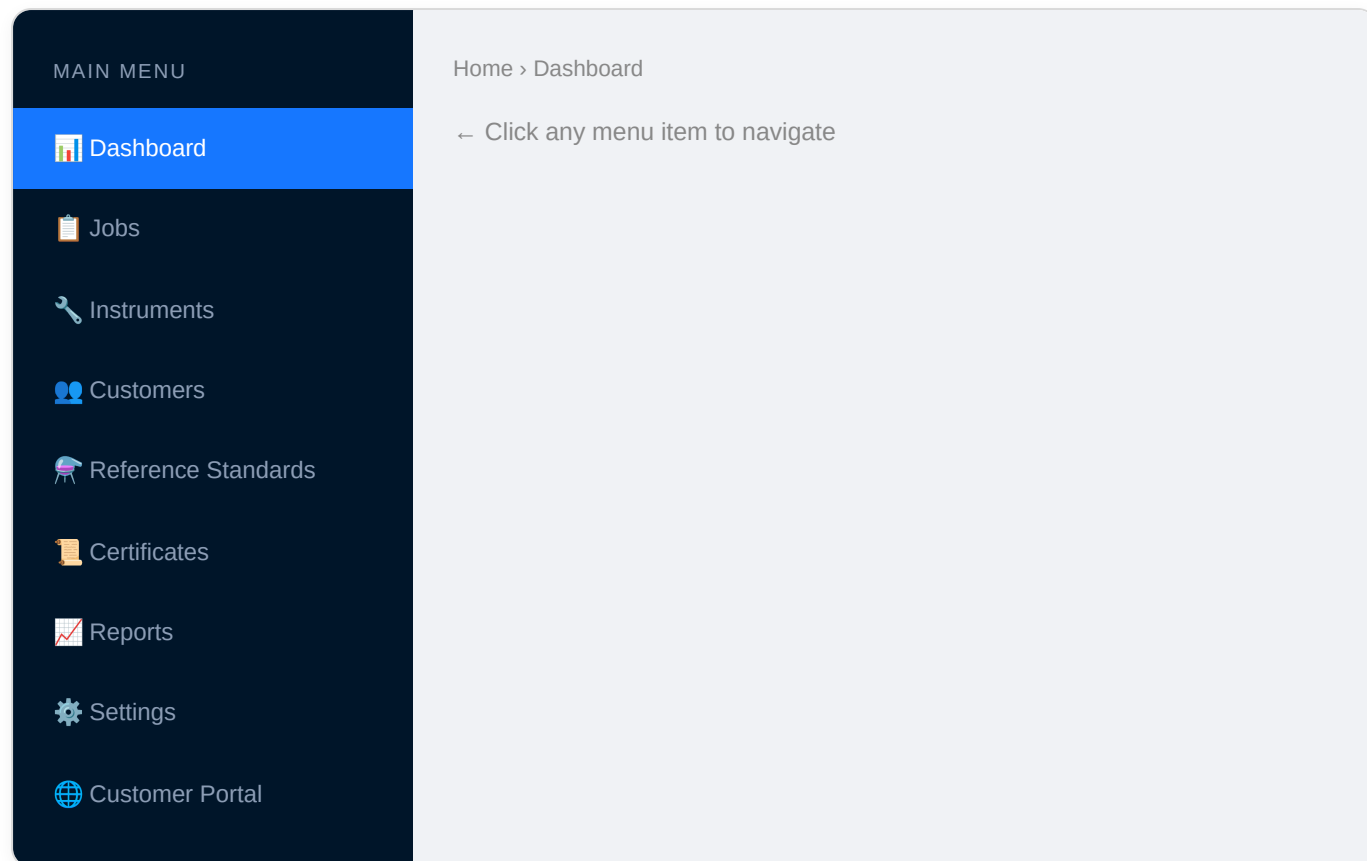
- 1 Enter your **Email Address** (registered by your lab administrator).
- 2 Enter your **Password**.

3

Click **Sign In**. You will be taken to the Dashboard.

1.2 Main Navigation

After login, the left sidebar is always visible. Each item opens a different module.



 **Tip:** The sidebar can be collapsed on smaller screens by clicking the hamburger icon (≡) at the top left.

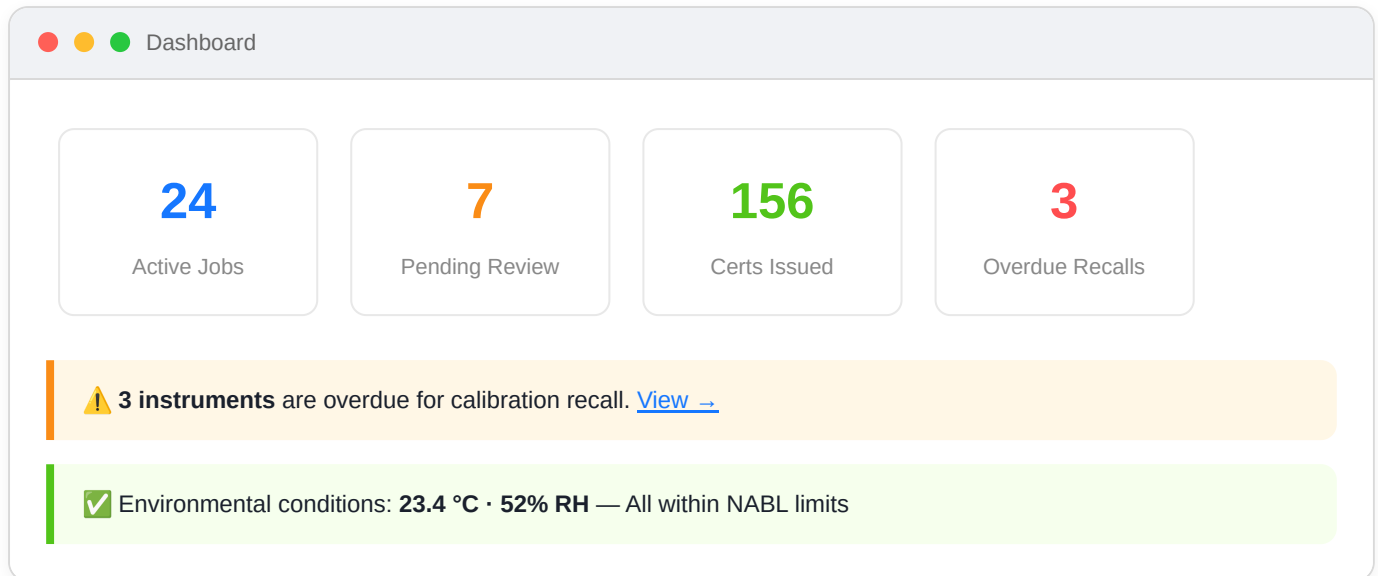
1.3 User Roles

Role	Access Level
ADMIN	Full access — manage users, settings, all jobs
ENGINEER	Create & work on calibration jobs assigned to them
REVIEWER	Review and approve calibration datasheets
TECHNICAL_MANAGER	Sign off technical approval
QUALITY_MANAGER	Final quality lock before certificate generation

2 Dashboard — At a Glance

Overview of lab activity, pending jobs, and calibration alerts

The Dashboard is the home screen. It shows real-time statistics and alerts so you know immediately what needs attention.



Key Dashboard Elements

- **Active Jobs** — Jobs currently in progress (not yet delivered or closed)
- **Pending Review** — Datasheets submitted and awaiting reviewer sign-off
- **Certs Issued** — Total calibration certificates generated to date
- **Overdue Recalls** — Customer instruments that have passed their due calibration date
- **Environmental Alert** — Live lab condition status (temperature 21–25 °C, humidity 35–65 %RH)

💡 Click any statistic card to navigate directly to the relevant module.

3 Jobs — Creating & Managing Calibration Jobs

Every calibration starts as a Job. This module handles intake, assignment, and tracking.

3.1 The Job List

Navigate to **Jobs** in the sidebar. You see a table of all calibration jobs with their current status.

Jobs

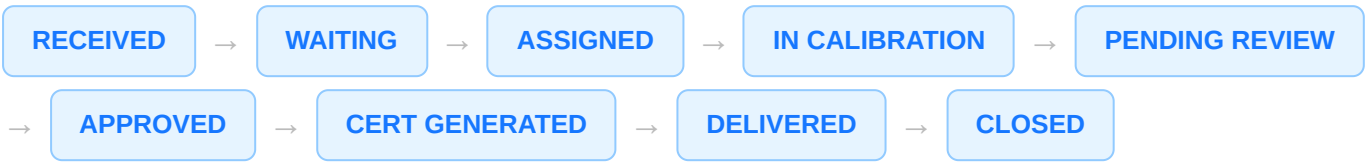
Search jobs...

Bulk Create Jobs

+ New Job

Job No.	Customer	Instrument	Received	Status	Engineer	Actions
JOB-2026-001	Tata Motors	Vernier Caliper	20 Jun 2026	IN CALIBRATION	Raj Kumar	Open
JOB-2026-002	BHEL	Pressure Gauge	21 Jun 2026	PENDING REVIEW	Anita S.	Open
JOB-2026-003	ISRO	Torque Wrench	22 Jun 2026	CERTIFICATE GENERATED	Priya M.	Open

3.2 Job Status Workflow



3.3 Creating a New Job

- 1 Click + **New Job** (top right of Jobs page).
- 2 Fill in the required details in the modal form.
- 3 Click **Create Job**. The job appears in the list with status **RECEIVED**.

Create New Job

Customer *

Tata Motors Ltd.

Instrument *

Vernier Caliper 300mm

Received At *

2026-06-22

PO Number

Optional

Challan No.

Condition of Item

Onsite Calibration?☐ Check if calibration is performed at customer's site

3.4 Bulk Job Creation

Use **Bulk Create Jobs** when a customer sends multiple instruments at once.

- 1 Click **Bulk Create Jobs** button.
- 2 Select the **Customer** from the dropdown.
- 3 A list of all instruments belonging to that customer appears. **Tick** the instruments to be calibrated in this batch.
- 4 Click **Create N Jobs**. One job is created per selected instrument simultaneously.

💡 Bulk creation is ideal for annual maintenance contracts (AMCs) where an entire fleet of instruments arrives together.

3.5 Searching & Filtering Jobs

Use the **search bar** at the top to filter jobs by job number, customer name, or instrument name in real time.

4 Job Workspace — Complete Calibration Workflow

The central workspace where engineers perform calibration, record data, and generate certificates

Click **Open** on any job to enter the Job Workspace. This is the most feature-rich screen in CLMS. It contains five tabs:

**Datasheet**

Uncertainty



Certificate



History



Remarks

4.1 Datasheet Tab

This is where the actual calibration measurements are entered. The datasheet records Before and After calibration readings for each measurement point.

4.1.1 Accuracy Class Selector

At the top of the Datasheet tab, select the instrument's **accuracy class**. This determines:

- **MPE (Maximum Permissible Error)** — The tolerance threshold for PASS/FAIL evaluation
- **Minimum Readings** required per point (per NABL 129 criteria)
- **Calibration Interval** recommendation in months

Datasheet Tab — Accuracy Class

 **NABL 129 Calibration Criteria**

Accuracy Class

Class I (±0.01 mm) ▼

Effective MPE

±0.01 mm

Min. Readings/Point

5 readings

Calibration Interval

12 months

4.1.2 Environmental Conditions

Record the laboratory environmental conditions at the time of calibration. NABL requires these to be within controlled limits.

Environmental Conditions

Temperature (°C)

23.4

Humidity (%RH)

52

Pressure (kPa)

101.3

☒ All environmental conditions are within NABL acceptable limits (Temp: 21–25°C, Humidity: 35–65%RH)

 If temperature or humidity is **outside** the NABL-specified range, the input field turns red and a warning alert appears. Do not proceed with calibration until conditions are within spec or document a deviation justification.

4.1.3 Measurement Readings

Enter the **Before** and **After** calibration readings for each nominal value point. The system automatically calculates:

- **Mean** of multiple readings per point
- **Error** (difference from nominal)
- **PASS / FAIL** against the selected MPE

Measurement Data Entry								
Nominal (mm)	R1	R2	R3	R4	R5	Mean	Error	Result
0.00	0.00	0.00	0.01	0.00	0.00	0.002	+0.002	PASS
10.00	10.01	10.01	10.00	10.01	10.01	10.008	+0.008	PASS
50.00	50.02	50.02	50.01	50.02	50.02	50.018	+0.018	FAIL

4.1.4 Workflow Actions

After entering all readings, use the action buttons to progress the job:

Button	Who Can Use	What It Does
Submit for Review	Engineer	Changes status to PENDING_REVIEW, notifies Reviewer
Approve	Reviewer / Technical Manager	Signs off the datasheet; progresses workflow
Request Correction	Reviewer	Sends job back to Engineer with comments
Generate Certificate	Quality Manager	Creates the locked calibration certificate

4.2 Uncertainty Tab

The Uncertainty tab implements a **GUM-compliant uncertainty budget** calculation. This is mandatory for NABL accredited laboratories.

- 1
- The system lists all **Uncertainty Sources** (Type A and Type B) relevant to this instrument type.
- 2
- For each source, enter or confirm the **value, distribution type** (Normal/Rectangular/U-shaped), and **divisor**.

- 3 The system calculates **Standard Uncertainty (u)** for each source.
- 4 **Combined Uncertainty (Uc)** is computed as the square root of the sum of squares.
- 5 **Expanded Uncertainty (U)** = $U_c \times \text{Coverage Factor } k$ (usually $k=2$ for 95% confidence).

Uncertainty Budget					
Source	Type	Value	Distribution	Divisor	Std. Unc.
Reference Std. Uncertainty	B	0.0025 mm	Normal	2	0.00125
Repeatability	A	0.003 mm	Normal	$\sqrt{5}$	0.00134
Resolution	B	0.005 mm	Rectangular	$\sqrt{3}$	0.00289
Expanded Uncertainty U (k=2, 95%)					0.0068 mm

4.3 Certificate Tab

Once the job is **APPROVED**, the Certificate tab becomes active. Here you can preview and generate the final calibration certificate.

YOUR LAB
logo here

CALIBRATION CERTIFICATE
Certificate No: CC-2026-0047

NABL¹⁴³

Instrument	Make/Model	Serial No.
Vernier Caliper 300mm	Mitutoyo / 530-104	12345678
Customer	Calibrated On	Next Due Date
Tata Motors Ltd.	22 Jun 2026	22 Jun 2027

This certificate is traceable to National / International Standards. Expanded Uncertainty: $U = 0.007 \text{ mm}$ ($k=2$, 95% confidence)

Dr. A. Kumar
Technical Manager

💡 Click **Download Certificate** to get a PDF-ready HTML version. Use your browser's print function (Ctrl+P) and select "Save as PDF" for a print-quality output.

4.4 Calibration History Tab

Shows all **previous calibration jobs** for the same instrument. This gives a complete longitudinal history — useful for trend analysis and NABL audits.

Job No.	Date	Engineer	Result	Certificate
JOB-2025-044	15 Jun 2025	Raj Kumar	PASS	CC-2025-0044
JOB-2024-031	12 Jun 2024	Anita S.	PASS	CC-2024-0031
JOB-2023-019	08 Jun 2023	Raj Kumar	PASS (Warning)	CC-2023-0019

4.5 Remarks Tab

A timestamped **communication log** for this job. Engineers, reviewers, and managers can add notes that are visible to all team members.

Remarks / Notes Timeline

Raj Kumar · 22 Jun 2026, 10:45 AM

Instrument received with minor surface rust on jaw faces. Cleaned before calibration. Condition documented with photos.

Anita S. (Reviewer) · 22 Jun 2026, 2:30 PM

Point 3 reading at 50mm shows marginal FAIL. Please re-check reference standard and repeat measurement.

Raj Kumar · 22 Jun 2026, 3:15 PM

Re-calibrated using freshly cleaned gauge blocks. Point 3 now PASS. Datasheet updated.

Add a remark...

Add

5 Instruments — Asset Registry

Manage all customer instruments received for calibration

Every instrument that comes into the lab for calibration must be registered here. Instruments are linked to customers and tracked across multiple calibration cycles.

5.1 Adding a New Instrument

- 1
- Go to **Instruments** in the sidebar.

- 2 Click + **Add Instrument**.
- 3 Fill in the instrument details and click **Save**.

Add Instrument

Instrument Name *		Customer *	
Vernier Caliper 300mm		Tata Motors Ltd.	
Make	Model	Serial Number	
Mitutoyo	530-104	12345678	
Range		Least Count / Resolution	
0–300 mm		0.02 mm	
Calibration Interval (months)		Asset Tag / ID	
12		Customer's internal tag	

5.2 Import Instruments from CSV

To add many instruments at once, use the **Import CSV** button. Download the template, fill it in Excel, and upload.

 **CSV columns for import:** Name, Make, Model, Serial Number, Range, Least Count, Customer Code, Calibration Interval Months

5.3 Calibration Due Tracking

After each calibration, the system automatically sets `lastCalibrationDate` and calculates `nextDueDate` based on the calibration interval. The instrument list shows a colour-coded status:

- OK** — Due date more than 30 days away
- Due Soon** — Due within 30 days
- Overdue** — Due date has passed

6 Customers — Client Management

6.1 Adding a Customer

- 1 Go to **Customers** in the sidebar.
- 2 Click **+ Add Customer**.
- 3 Enter all contact details and click **Save**.

Customer Details

Company Name *

Tata Motors Ltd.

Customer Code *

TML-001

Contact Email

quality@tatamotors.com

Phone

+91 98765 43210

Address

Plot 42, MIDC Industrial Area, Pune - 411018

GSTIN

Optional — for invoice records

6.2 Portal Access

Each customer can access the **Customer Portal** (see Section 11) using their registered email and your lab's accreditation code. The portal lets them track jobs and download certificates without calling you.

 Share your lab's accreditation code with customers when onboarding them. This acts as a shared secret for portal login.

Reference Standards are your lab's own calibrated master instruments used to calibrate customer instruments. NABL requires all masters to be traceable to national/international standards and within their calibration validity period.

Reference Standards

12
Total Standards

2
Due Soon (30d)

1
Overdue

1 reference standard(s) are OVERDUE for calibration. Overdue reference standards must not be used for calibration. Schedule immediate recalibration to maintain NABL accreditation.

Standard	Make/Model	Traceability	Cert No.	Calibration Due	Status
Gauge Block Set 1mm ID: STD-001 · SN: GB-2019-001	Mitutoyo / 516-XXX	NPL India	NPL/2025/001	15 Dec 2026	Valid
Load Cell 50kN ID: STD-005 · SN: LC-5001	HBM / U2B	NABL Lab	NABL/25/0445	01 Jul 2026	Due Soon
Pressure Deadweight ID: STD-008 · SN: DW-3322	Budenberg / —	NPL India	NPL/2025/088	10 May 2026	OVERDUE

7.1 Adding a Reference Standard

- 1 Click **Add Standard**.
- 2 Enter: Standard Name, ID Number, Make, Model, Serial Number, Traceability, Certificate Number, Uncertainty, Calibrated Date, Calibration Due Date, Location.
- 3 Click **Add** to save.

7.2 Alerts

- **Red banner:** One or more standards are OVERDUE — do not use for calibration

- **Orange banner:** Standards due within 30 days — schedule recalibration

 Using an OVERDUE reference standard for calibration is a major non-conformance under NABL criteria and could result in suspension of your accreditation. Always check the Reference Standards page before starting calibration work.

7.3 Export & Import

Use **Export CSV** to download the full master list. Use **Import CSV** to bulk-upload standards from a spreadsheet.

8 Certificates — View & Download

Access all generated calibration certificates in one place

The Certificates module lists all locked (finalised) calibration certificates. Once generated, a certificate cannot be edited — it is permanently locked to maintain document integrity.

Certificates					
Cert. No.	Job No.	Customer	Instrument	Issue Date	Actions
CC-2026-0047	JOB-2026-001	Tata Motors	Vernier Caliper 300mm	22 Jun 2026	Download
CC-2026-0046	JOB-2026-002	BHEL	Pressure Gauge 0-10 bar	20 Jun 2026	Download

Downloading a Certificate

- 1 Find the certificate in the list (search by cert number, customer, or instrument).
- 2 Click **Download**. The certificate opens in a new browser tab as a formatted HTML page.
- 3 Press **Ctrl+P** (Windows) or **Cmd+P** (Mac), choose **Save as PDF** as the destination.
- 4 Save and share the PDF with your customer.

💡 Customers can also download their own certificates directly from the Customer Portal (Section 11) without needing to contact you.

9 Reports — Analytics & Exports

Calibration recall reminders, job register, engineer performance, and data export

The Reports module has four tabs. Navigate to **Reports** in the sidebar.

9.1 Calibration Recall Report

Shows customer instruments that are **overdue** or **due soon** for recalibration. Use this to proactively contact customers before their instruments expire.

Reports — Calibration Recall			
Show instruments due within: 30 days ▼			
Instrument	Customer	Next Due	Interval
Pressure Gauge 0-10 bar Yokogawa · SN P-2201	BHEL quality@bhel.in	01 Jul 2026 in 5d	12 months
Torque Wrench 10-100 Nm Snap-on · SN TW-0099	ISRO	10 May 2026 OVERDUE	6 months

Change the **window** (7, 30, 60, or 90 days) to see more or fewer instruments.

9.2 Job Register

A complete list of all calibration jobs — useful for NABL audits and management review. Shows job number, date, customer, instrument, job type (in-lab / onsite), and status.

9.3 Engineer Performance

An analytical view of each engineer's workload and completion metrics.

48

Total Jobs

36

Assigned

29

Completed

3

Engineers Active

Engineer	Total Jobs	Completed	Completion Rate	Avg. Turnaround
Raj Kumar	18	15	83%	4 days
Anita Sharma	12	10	83%	3 days
Priya Mehta	6	4	67%	9 days

9.4 Lab Data Export

Export raw data for record-keeping, audits, and external analysis. Select a time period (last 1 or 2 years) and choose what to export:

Export Jobs

All jobs with customer, instrument, status, and creation date.

[Export Jobs CSV](#)

Export Certificates

All issued certificates with certificate number and details.

[Export Certificates CSV](#)

Export Environmental

All temperature, humidity, and pressure records.

[Export Environmental CSV](#)

Full Lab Report

Combined jobs + customers in one CSV file.

[Export Full Report CSV](#)

10 Settings — Lab Configuration

Configure your lab profile, certificate appearance, uncertainty types, and user management

Go to **Settings** in the sidebar. Only Admin users can access most settings.

10.1 Lab Profile

Set your laboratory's identity information. This appears on every certificate generated by CLMS.

Settings — Lab Profile

Lab Name *

National Calibration Laboratory

NABL Accreditation No.

CC-1234

Phone

+91 20 2765 4321

Email

lab@ncl.in

Website

www.ncl.in

Address

Plot 12, Industrial Estate, Pune - 411019

Lab Logo URL

https://cdn.yourlab.com/logo.png

Certificate Signature Block

Authorized Signatory Name

Dr. A. Kumar

Signatory Designation

Technical Manager

Appears on all certificates

Save Lab Details

10.2 Certificate Templates

CLMS supports multiple certificate layout templates. Select the one that matches your NABL scope and preferred format.

10.3 Uncertainty Types

Define the uncertainty budget categories (Type A / Type B sources) that appear in the uncertainty calculation for each instrument type. These are configurable per discipline (dimensional, pressure, electrical, etc.).

10.4 User Management

Admins can add, edit, and deactivate user accounts. When adding a user, assign their role carefully — it controls what they can see and do in the system.

1

Go to Settings → **Users** tab.

- 2 Click **Add User**.
- 3 Enter their name, email, and select a **role**.
- 4 The user receives a welcome email with instructions to set their password.

11 Customer Portal — Self-Service Access

Your customers can track their jobs and download certificates without contacting you

The Customer Portal is a **separate login screen** accessible via the **Customer Portal** menu item or by navigating to `/portal` in the URL. It uses a different login — customers do NOT use the main CLMS login.

11.1 Customer Login

Customer Portal — Login



Customer Portal

Track your calibration jobs & certificates

Email Address

Lab Accreditation Code

Access Portal

Customers log in using:

- **Email Address** — The email registered for their company in CLMS
- **Lab Accreditation Code** — Your NABL accreditation number (e.g., CC-1234). Share this with customers when you onboard them.

11.2 Portal — Jobs Tab

After login, customers see their **active and completed jobs**.

Customer Portal — My Jobs				
Job No.	Instrument	Status	Received	Certificate
JOB-2026-001	Vernier Caliper 300mm	IN CALIBRATION	20 Jun 2026	Pending
JOB-2025-044	Vernier Caliper 300mm	DELIVERED	15 Jun 2025	✓ CC-2025-0044

11.3 Portal — Certificates Tab

Customers can see all their **issued certificates** and download them directly.

- 1 Click the **Certificates** tab.
- 2 Find the certificate by instrument or date.
- 3 Click **Download**. The certificate opens as HTML in a new tab.
- 4 Use `Ctrl+P` → Save as PDF.

11.4 Portal — Instruments Tab

A dashboard of all customer instruments with calibration due date status, and a **View History** button to see the full calibration timeline for each instrument.

Customer Portal — My Instruments					
Name	Serial No.	Last Calibrated	Next Due	Status	History
Vernier Caliper 300mm	12345678	22 Jun 2026	22 Jun 2027	OK	View History
Pressure Gauge 0-10 bar	P-2201	15 Jan 2026	15 Jul 2026	Due Soon	View History

17

Calibration History — Vernier Caliper 300mm

Job No.	Received	Status	Certificate
JOB-2026-001	20 Jun 2026	IN CALIBRATION	—
JOB-2025-044	15 Jun 2025	DELIVERED	CC-2025-0044

12 Role Permissions Summary

Quick reference for what each role can and cannot do

Feature	ADMIN	ENGINEER	REVIEWER	TECH MGR	QA MGR
Create Job	✓	✓	✓	✓	✓
Enter Datasheet Readings	✓	✓	—	—	—
Submit for Review	✓	✓	—	—	—
Approve / Request Correction	✓	—	✓	✓	—
Generate Certificate	✓	—	—	—	✓
Manage Reference Standards	✓	✓	—	✓	—
View Reports	✓	✓	✓	✓	✓
Manage Settings / Users	✓	—	—	—	—

💡 **Need help?** Contact your lab administrator or refer to the in-app tooltips (hover over any field with a ? icon).



Calibration Laboratory Management System

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